GABRIELE BOZZOLA *PhD candidate, Department of Astronomy, University of Arizona, United States*

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IN A NUTSHELL

- Computational physicist and software developer specialized in high-performance and scientific computing.
- Experienced Python programmer, author of kuibit, the analysis and visualization library for Einstein Toolkit.
- Prolific scientist, with 10+ publications, 100+ citations, 150k$+ in grants, 2 awards for exceptional research.
- Passionate instructor and communicator, with 30+ invited/contributed talks, 1 award for excellence in teaching.
- Expert in large-scale simulations, with 10M+ CPU-hours awarded on national supercomputers.
- Active member of the open-source world, maintainer of vterm–the fastest terminal for Emacs (1.5k+ GitHub stars).

INTERESTS

Numerical relativity, strong gravity, black holes, neutron stars, accretion disks, high-performance computing, software engineering, open source software, open science, community and inclusion.

EDUCATION

2018–2023	University of Arizona — MSc and PhD in Astronomy and Astrophysics (GPA: 4/4, Advisor: Prof. Vasileios Paschalidis, expected PhD defense: May 2023)
2015–2017	University of Milan — Master in Physics (*cum laude*, GPA: 30/30, Advisor: Prof. Nikolaos Stergioulas)
2012–2015	University of Milan — Bachelor in Physics (*cum laude*, GPA: 29.35/30, Advisor: Prof. Valentino Liberali)

RELEVANT WORKING EXPERIENCE

2018–2023	Graduate Research and Teaching Assistant at the University of Arizona, Tucson (AZ)
2016	Part-time Student Linux System Administrator at Multimedia and Computation Lab, Milan (IT)

PEER-REVIEWED PUBLICATIONS (9 total; 8 first author, 1 second author; 2 single author; 2 letters)

9. R. Luna, **G. Bozzola**, V. Cardoso, V. Paschalidis, M. Zilhão, *Kicks in charged black hole binaries*, PRD 106, 8, 084017 (2022)
8. **G. Bozzola**, C. Chan, V. Paschalidis, *Black Hole Physics and Computer Graphics*, CISE 24, 2 (2022)
7. **G. Bozzola**, *Does charge matter in high-energy collisions of black holes?*, PRL 128, 4, 071101 (2022)
6. **G. Bozzola**, V. Paschalidis, *Numerical-relativity simulations of the quasi-circular inspiral and merger of non-spinning, charged black holes: methods and comparison with approximate approaches*, PRD 104, 4, 044004 (2021)
5. **G. Bozzola**, *kuibit: Analyzing Einstein Toolkit simulations with Python*, JOSS 60(6), 3099 (2021)
4. **G. Bozzola**, V. Paschalidis, *General relativistic simulations of the quasi-circular inspiral and merger of charged black holes: GW150914 and fundamental physics implications*, PRL 126, 041103, (2021)
3. **G. Bozzola**, P. Espino, C. Lewin, V. Paschalidis, *Maximum mass, solution space and universal relations of rotating relativistic hadron-quark hybrid stars*, EPJ A 55 9, 149, (2019)
2. **G. Bozzola**, V. Paschalidis, *Initial data for general relativistic simulations of multiple electrically charged black holes with linear and angular momenta*, PRD 99, 10, 104044 (2019)
1. **G. Bozzola**, N. Stergioulas, A. Bauswein, *Universal relations for differentially rotating relativistic stars at the threshold to collapse*, MNRAS 474, 3557–3564 (2017)

PAPERS UNDER REVIEW (1 total; 1 second author)

1. P. Espino, **G. Bozzola**, V. Paschalidis *Quantifying uncertainties in general relativistic magnetohydrodynamic codes*, Submitted to PRD, arXiv: 2210.13481

CONFERENCE PAPERS (2 first author)

2. **G. Bozzola**, V. Paschalidis, *Generation of Initial Data for General-Relativistic Simulations of Charged Black Holes*, in "Einstein Equations: Physical and Mathematical Aspects of General Relativity", Birkhäuser (2019)
1. **G. Bozzola**, L. Frontini, V. Liberali, S. Shojaii, A. Sabile, *Improvement of radiation tolerance in CMOS ICs through layout-oriented simulation*, in "5th International Conference on Modern Circuits and Systems Technologies (MOCAST)", IEEE (2016)

WHITEPAPERS

1. E. Barausse et al. (with **G. Bozzola**), *Prospects for Fundamental Physics with LISA*, GR&Grav 52, 81 (2020)

SOFTWARE (3 total (published); 1 lead developer)

3. Einstein Toolkit maintainers et al. (with **G. Bozzola**), *The Einstein Toolkit*, Zenodo, 3350841 (2021 and later)
2. H. Witek, M. Zilhão, **G. Bozzola**, et al., *Canuda*, Zenodo, 3565474 (2021 and later)
1. **G. Bozzola**, *kuibit*, Zenodo, 4679862 (2021 and later)

AWARDS

2022	Distinguished Student Award, Forum on International Physics, American Physical Society
2022	Graduate Student Research Prize, Theoretical Astrophysics Program, University of Arizona
2022	Department of Astronomy award for excellence in Teaching, University of Arizona

SUPERCOMPUTING TIME (AS PI OR CO-I)

2021–2023	146,250 SBUs on NASA's High-End Computing Center (HECC) (equivalent to $ 68,738)
2020	50,000 SUs on Frontera at Texas Advanced Computing Center (TACC)

GRANTS

2020–2023	NASA Future Investigator in NASA Earth and Space Science ($ 105,000)
2020–2021	TACC Frontera Fellowship ($ 44,000)
2017–	Travel grants from NewCompStar, American Physical Society (APS) divisions, University of Arizona Theoretical Astrophysics Program (TAP), Graduate Student and Professional Council (GSPC), National Science Foundation (NSF), Institute of Theoretical Physics at the Goethe University (total: $ 9,500)

TEACHING

Fall 2022	Guest lecture on gravitational waves in ASTRO300A (main instructor: A. Youdin)
Fall 2022	*Introduction to the Einstein Toolkit*, series of seminars at the University of Arizona
Spring 2022	TA for PHYS105A: *Introduction to Scientific Computing* (main instructor: C. Chan)
Spring 2022	Physics consultation room (academic support for undergraduate physics courses)
Fall 2021	TA for ASTR300A: *Dynamics and Mechanics in Astrophysics* (main instructor: A. Youdin)
Fall 2021	Mentor for Astronomy Tutoring for Majors & Minors Program (ATOMM)
Spring 2017	*Computational tools for physicists*, series of seminars at the Aristotle University of Thessaloniki (GR)

MENTORING

Graduate, University of Arizona (main advisor: V. Paschalidis):
Margaret Smith (2021–), Mahdi Naseri (2022–)

Undergraduate, University of Arizona (main advisor: V. Paschalidis):
Collin Lewin (2018–2019), William Lake (2018–2019), Cyrus Worely (2022–)

INVITED TALKS AND SEMINARS (18 in total)

2023	*Introduction to the `Einstein Toolkit`*, Instituto Superior Tecnico (PT)
2023	*The high energy collisions of (charged) black holes*, Instituto Superior Tecnico (PT)
2023	*The high energy collisions of (charged) black holes*, University of Valencia (ES)
2023	*The high energy collisions of (charged) black holes*, Goethe University in Frankfurt (DE)
2022	*Nonlinear simulations of charged binary black holes as a tool for theoretical physics*, Perimeter Institute (virtual)
2022	*Exploring nonlinear interactions between charged black holes with numerical relativity: gravitational waves and fundamental physics*, KIPAC Tea talks, Stanford (virtual)
2022	*Advancements in fundamental physics with numerical relativity simulations of charged black holes*, Theoretical Astrophysics Program Colloquia, University of Arizona (AZ)
2022	*Advancements in fundamental physics with numerical relativity simulations of charged black holes*, University of Milan-Bicocca, Milan (IT)
2022	*Post-processing Cactus simulations with Python*, Einstein Toolkit North American School, University of Idaho (ID)
2022	*Advanced tools for Python projects*, LCM (Multimedia and Computing Lab) in Milan (IT)

2021 *A brief introduction to the Journal of Open-Source Software*, Theoretical Astrophysics Program, Computation&Data Initiative, University of Arizona (AZ)

2021 *kuibit: Analyzing Einstein Toolkit simulations with Python*, Einstein Toolkit North American School (virtual)

2021 *Ray-tracing and radiation-transfer in dynamical spacetimes*, Texas Advanced Center for Computing (virtual)

2021 *Exploring the future of supercomputing*, IBM EMEA Project Managers workshop (virtual)

2021 *kuibit: a community-friendly tool to analyze Cactus simulations*, Einstein Toolkit seminars (virtual)

2020 *Numerical relativity simulations of charged black holes*, Los Alamos National Laboratory (virtual)

2017 *Computational tools for physicists*, series of seminars, Aristotle University of Thessaloniki (GR)

2016 *Introduction to GNU/Linux*, series of seminars, LCM (Multimedia and Computing Lab) in Milan (IT)

CONTRIBUTED TALKS (11 in total)

2022 *Fundamental physics probed via high-energy collisions of electrically charged black holes*, American Physical Society April Meeting in New York City (NY)

2022 *Fundamental physics probed via high-energy collisions of electrically charged black holes*, Midwest Relativity Meeting (virtual)

2021 *General-relativistic simulations of quasi-circular inspirals of charged black holes*, American Physical Society April Meeting (virtual)

2021 *telega.el and the Emacs community on Telegram*, EmacsConf 2021 (virtual)

2020 *Advancements in general-relativistic simulations of charged black holes*, Pacific Coast Gravity Meeting (virtual)

2020 *Are black holes charged?*, Early Career Scientist Talks at Steward Observatory (virtual)

2020 *A tour of vterm*, EmacsConf 2020 (virtual)

2019 *Initial data for general-relativistic simulations of generic black hole systems with electric charge, linear and angular momenta*, American Physical Society April Meeting in Denver (CO)

2018 *Initial data for general-relativistic simulations of generic black hole systems with electric charge, linear and angular momenta*, Alpine School for Theoretical Physics in Domodossola (IT)

2018 *Quasi-radial instability of differentially rotating relativistic stars*, American Physical Society April Meeting in Columbus (OH)

2017 *Stability and universal relations of differentially rotating neutron stars*, NewCompStar Summer School in Sofia (BG)

SERVICE

2020– Building, maintaining, and providing support for software for the scientific community (mainly, kuibit) and for the open-source community (mainly, emacs-libvterm)

2020– Contributing to the public codes Einstein Toolkit and canuda

2020– Referee for *Physics Review D*, *Science Bulletin*, *Journal of Open Source Software*

COMMUNITY, DIVERSITY, EQUITY, INCLUSION

2022 Graduate student representative for Steward Observatory graduate school admissions

2020– Founding member of Steward Observatory International Scholars Task Force

2020– Creating and supporting software to lower the entry-barrier to numerical relativity

2019 Outreach activities with K-12 classes at the Planetarium of the University of Arizona

PROFESSIONAL MEMBERSHIPS

2018– LISA (Laser Interferometer Space Antenna) Science Group

2018– TAP (Theoretical Astrophysics Program) at the University of Arizona

COMPUTER SKILLS (working experience)

Python, C, C++ (pre C++11), Fortran, Emacs Lisp, Bash, LaTeX, awk, OpenMP, HTCondor, SLRUM, HDF5, regexp, High-Performance Computing, GNU/Linux system administration, git, benchmarking, continuous development and integration, computational methods, algorithm development.

CONTRIBUTIONS TO OPEN-SOURCE SOFTWARE

Lead developer:

kuibit	Post-processing and visualization Python package for `Einstein Toolkit`.
motionpicture	Effortlessly make animations with Python.
ivy-emoji	Insert emojis in Emacs buffers.
vterm-extra	Additional features for Emacs's vterm.
gray-python	Tools for the GRay2 general-relativistic ray-tracing code.
sphinx-citations	Fetch citations from ADS and add them to Sphinx-generated pages.
Jhuki	Preparing simulations for `Einstein Toolkit`.
ciak	Orchestrate running scripts.

Active contributor:

Einstein Toolkit	Relativistic astrophysics, gravitational-wave astronomy, and numerical relativity simulations.
Canuda	Fundamental physics with numerical relativity.
vterm	The best terminal emulator for Emacs. (Maintainer, with libvterm-mirror)

LANGUAGES AND NATIONALITY

Italian (mother tongue), English (full proficiency); Italian nationality.

REFERENCES

1. Vasileios Paschalidis, Assistant Professor at the Astronomy and Physics Departments, University of Arizona (US), vpaschal@arizona.edu
2. Chi-kwan Chan, Associate Research Professor at the Astronomy Department and Affiliate Member of the Applied Mathematics Department, University of Arizona (US), chanc@arizona.edu
3. Dimitrios Psaltis, Professor at the Physics Department, Georgia Tech (US), dpsaltis@gatech.edu
4. Nikolaos Stergioulas, Professor and Director of the Astronomy Department, Aristotle University of Thessaloniki (GR), niksterg@auth.gr

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